

National Entrepreneurship Challenge (NEC)

E-Cell, IIT Bombay

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Team: Titan - Agritech & Drones

My Motivation

Growing up visiting my grandparents' farm in Telangana, I saw firsthand how sudden pest outbreaks could wipe out an entire season's cotton crop in days. Manual pesticide spraying is not only slow and uneven, but it exposes farm workers to hazardous chemicals without proper protective gear.

I started building custom payload drones designed specifically for small, fragmented landholdings. Building agricultural drones isn't just about flying multirotors; it's about flight battery efficiency, spray nozzle calibration, and ruggedizing electronics to withstand dust and heat. My motivation is to bring precision farming tools to smallholder farmers at an affordable service cost.

My Expectations from E-Cell

The drone sector in India involves complex DGCA airspace regulations, Type Certification, and pilot licensing mandates. I expect E-Cell to offer guidance on regulatory compliance, aviation insurance, and IP protection.

Through NEC, I want to evaluate different business models--specifically comparing direct hardware sales against a 'Drone-as-a-Service' (DaaS) model operated through local farmer cooperatives. I want to build a resilient operational strategy before expanding field pilots.